



Alteration of compacted bentonite adjacent to a metal canister under continuous real groundwater flow: A 6.5-year study in Korea and its implications for repository safety

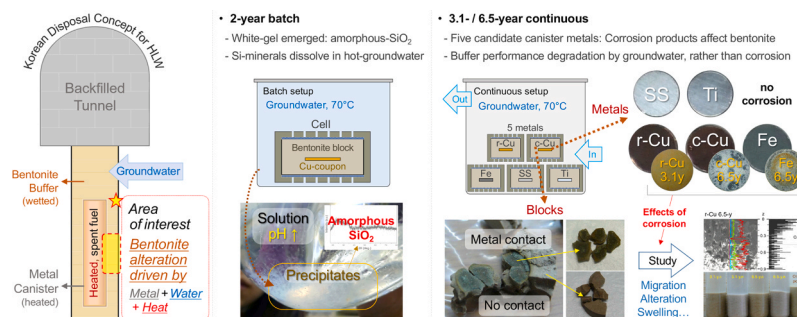
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HIGHLIGHTS

- Long-term bentonite alteration was studied for 6.5 years using KURT groundwater at 70°C.
- Mineral dissolution was mainly driven by groundwater, with limited influence from metal corrosion.
- Ca-precipitates formed only on corroded metals, indicating precipitation driven by cations from bentonite.
- Illitization was not detected but cannot be excluded over longer periods under KURT groundwater conditions.
- Buffer degradation affecting HLW sorption was driven mainly by groundwater leaching rather than metal corrosion.

GRAPHICAL ABSTRACT



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ABSTRACT

This study presents an intermediate stage of an ongoing long-term project. A 2-year batch setup and a continuous-flow setup (3.1- and 6.5-year) were conducted to investigate buffer alteration, employing cells designed to simulate the metal-bentonite buffer interface. These experiments were performed using real groundwater at 70°C, replicating the environment expected in actual repositories. In the batch setup, the observed pH trend and amorphous SiO₂ gel formation suggested a sequential mechanism where feldspar dissolution increases the pH to subsequently promote SiO₂ mineral dissolution, indicating a negligible role of metals in bentonite loss. In the continuous-flow setup involving five metal specimens (rolled and cold-sprayed Cu, Fe, SS, and Ti), investigations focused on: 1) the species and formation mechanisms of precipitates adhered on metals; 2) the corrosion mechanism and penetration of corrosion products; 3) the evaluation of apparent diffusivity for Cu²⁺; 4) mineral alterations in the bentonite; and 5) changes in buffer performance, specifically swelling index and dry density. Notably, adhered precipitates formed exclusively on corroded Cu and Fe specimens, indicating the participation of bentonite-derived Ca²⁺ in interfacial reactions. Furthermore, the inconsistency between corrosion amount and penetration distance for rolled and cold-sprayed Cu implied differences in corrosion modes depending on copper purity. Regarding buffer performance metrics, observed slight degradations were attributed primarily to groundwater-induced mineral leaching rather than corrosion. Given the spatiotemporal scales of the repository safety, these results provide a critical geochemical basis for predicting

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the fate of nuclear wastes, necessitating sustained monitoring to validate the long-term performance against radionuclide migration.

1. Introduction

The permanent isolation of high-level radioactive waste is a critical imperative arising from nuclear energy generation and a fundamental duty for humanity. Aligning with fundamental principles shared by disposal programs worldwide, the Korea Atomic Energy Research Institute (KAERI) has proposed the Korean Reference disposal System (KRS) as a national solution for deep geological disposal. The KRS ensures safety for one million years by adopting a multi-barrier approach that integrates an Engineered Barrier System (EBS) with host rock at a depth of over 500 m [1–4].

The bentonite buffer, a core component of the EBS, protects the canister by performing critical functions such as heat dissipation, delaying groundwater infiltration, mitigating external impacts, and retarding radionuclide migration [5–7]. From an engineering perspective, key parameters are defined, including thermal conductivity, swelling pressure, hydraulic conductivity, and cation exchange capacity (CEC). Although detailed specifications may vary by country, typical design criteria for these parameters ($\geq 1 \text{ W} \cdot \text{m}^{-1} \cdot \text{K}^{-1}$, $\geq 2.0 \text{ MPa}$, $\leq 10^{-12} \text{ m} \cdot \text{s}^{-1}$, and $\geq 0.06 \text{ meq} \cdot \text{g}^{-1}$) have been experimentally demonstrated to be closely correlated with the dry density of the bentonite; consequently, most countries adopt a target dry density of approximately $1.6 \text{ g} \cdot \text{cm}^{-3}$ as a primary design standard. It is critical to note that this design basis is premised on no chemical alteration of bentonite [8–12].

In the repository, the buffer is installed in a configuration where the bentonite block directly contacts and surrounds the metal canister. At this canister-bentonite interface, the synergistic action of the following factors, potentially degrading buffer performance, distinguishes this zone from the bulk bentonite: 1) heat from the spent fuel, 2) groundwater and dissolved substances, 3) oxygen from construction or via groundwater, 4) intrinsic materials within the bentonite, and 5) the canister itself. Considering the million-year timescale of the safety case assessment, the initial thermal period—roughly the first few decades—is expected to have a large contribution to the overall long-term behavior; hence, examining this interface on a scale of several years to decades is a research priority for research groups worldwide [7,13–17].

The alteration, driven by metal-involving reactions, mainly corrosion, for which the aforementioned factors are primarily responsible, is of primary interest; copper is a leading candidate material owing to its high corrosion resistance, and it is essential to understand how copper affects bentonite over the long term within the KRS. Disposal-related researchers have extensively studied on corrosion reactions and their products, including dissolved Cu^{2+} , dissolved CuCl_2^- , precipitated Cu_2O , precipitated $\text{CuCl}_2 \cdot 3\text{Cu}(\text{OH})_2$, and precipitated $\text{Cu}_2\text{S}/\text{CuS}$; (it is reasonable to treat CuO as a secondary product) [13,15,18–21]. However, from the perspective of the buffer bentonite, the physicochemical mechanisms by which these products exert their impacts require further study.

For solid-phase precipitates, an immediate chemical reaction with bentonite might be rare; instead, they may physically occupy the pore space, thereby affecting swelling pressure or hydraulic conductivity [22–24]. For aqueous-phase products, Cu^{2+} in particular is known to easily interact with bentonite; this interaction is rapid, features difficult desorption, and shows a higher affinity than Ca^{2+} . The most predictable mechanism is therefore an ion exchange for the candidate Ca-bentonite, resulting a Cu-bentonite [25,26]. Additionally, atomic diffusion enabling copper to penetrate the bentonite structure has also been reported [27,28]; elucidating these mechanisms remains an important mission, but is beyond the scope of this study.

This study is a year-scale observation premised on the possibility of bentonite alterations in the presence of metals, focusing primarily on Cu, using real groundwater. The experiment involved two setups: 1) A batch

setup maintained at 70°C for 2 years, and 2) A continuous flow setup where cells with five metal specimens (rolled, and cold-sprayed Cu, Fe, SS, and Ti), installed in 70°C chambers within the KURT (KAERI Underground Research Tunnel), were continuously supplied with real groundwater from 150 m below; separate 3.1-year and 6.5-year chambers were prepared.

The batch experiment focused on solution evolution, observing a pH increase over two years and the formation of a gel-like white amorphous silica. Based on this observation, a mechanism of mineral dissolution was proposed, which was found to be unrelated to metal-induced reactions. In the continuous configuration, a comprehensive analysis was performed on bentonite-derived materials on the metal surface, the metal-contact surface of the block, and the propagation of metal-derived species into the bulk bentonite. Based on visual observation and metal mass loss from corrosion, along with phenomena observed via Energy-Dispersive X-ray Spectroscopy (EDS), Inductively Coupled Plasma Optical Emission Spectroscopy (ICP-OES), and X-ray Diffraction (XRD) analyses, a governing corrosion mechanism was proposed, and the characteristics of Cu diffusion and propagation were evaluated. Furthermore, their presumed causes and related-mechanisms were discussed in detail, including: 1) the reactions related to releasing Ca^{2+} from bentonite; 2) differences in behavior between the copper types; and 3) the possibility of montmorillonite and other mineral alteration. Subsequently, key buffer performance parameters, the swelling index and block density, were evaluated to assess whether the bentonite was affected by these alterations.

The significance of this study lies in its use of a continuous supply of real underground water to observe long-term exposure phenomena over a 6.5-year period. By integrating metal observations, this paper proposes a rational mechanism, providing a reasonable basis for the most likely reactions. Most importantly, this work is a part of an ongoing project for longer observation.

2. Materials and methods

2.1. Materials

Ca-type bentonite, obtained at a mine in Yangnam-myeon, Gyeongju-si, Republic of Korea (named as KJ-I and KJ-II depending on the time of mining), was provided as powder (CLARIANT KOREA, Republic of Korea). KJ-I bentonite was the primary material used in the entire experiments. Its detailed mineralogical and chemical compositions, determined by X-ray diffraction (XRD) and X-ray fluorescence (XRF), are presented in Fig. S1, and Tables S1 and S2. KJ-II bentonite was used for simple comparison.

The bentonite blocks were fabricated via uniaxial compaction using bentonite powders with an initial water content of approximately 11–12 wt%. The blocks were compacted to an initial dry density of $1.72\text{--}1.73 \text{ g} \cdot \text{cm}^{-3}$ with a total dry bentonite mass of 22.337 g. This dry mass corresponds to a target dry density of $1.6 \text{ g} \cdot \text{cm}^{-3}$ based on the total internal volume of the rigid cell (13.960 cm^3), which represents the repository design standard. The higher initial dry density resulted from the blocks being intentionally fabricated with a slightly smaller size to facilitate assembly (Fig. S2). Additional details are provided in the Supplementary Information (SI).

As candidates for canister materials and as samples to simulate their role in the experiment, the following materials were individually prepared as coupon-shaped specimens (15 mm diameter; 1 mm thickness, total area 400 mm^2): rolled copper (r-Cu; off-the-shelf), cold-spray coated copper (c-Cu; fabricated, coated on cast-iron at 600°C main-gas, 400°C powder-preheat, and 30 bar; a gradient was Changseong copper

powder with 99.5 wt% Cu, 0.41 wt% O), Type 304 stainless steel (SS; off-the-shelf), titanium (Ti-G2; off-the-shelf), and nodular graphite cast iron (ng-Fe; fabricated, containing 2.5–5.0 wt% of C). Specifically, the copper products were supplied by JinYoung MT (Republic of Korea). The titanium (Ti-G2) and nodular graphite cast iron (ng-Fe) were obtained from Steelmax (Republic of Korea), while the Type 304 stainless steel (SS) was purchased from Si-Heung ST Co., Ltd. (Republic of Korea). All the metal-coupons were polished using 1 μm of diamond powder.

Calcium chloride (CaCl_2 , >99.5%, J.T. Baker), refined mineral oil (Shell Heat Transfer Oil S2, ISO VG32 grade, Shell), and deionized water (> 18.0 M Ω) were utilized to measure the dry density and swelling index of the compacted bentonite blocks retrieved after the 3.1- and 6.5-year exposure tests.

2.2. KURT groundwater: experimental conditions and reference data

Groundwater supplied to the chambers was sourced from a borehole at approximately 150 m depth in the RG-1 gallery of the KURT. Considering the starting and retrieval time points of the 3.1-year and 6.5-year experiments, groundwater chemistry data from 2010, 2014, and 2017 are considered most relevant. It is noted that, given the extended duration of the experiments and practical factors such as changes in experimental personnel, groundwater sampling could not be performed in a fully routine and time-resolved manner throughout the entire exposure period; this is acknowledged as a limitation of the study.

As a complementary approach to address this limitation, groundwater chemistry data measured at the end of the experiments in April 2017 and one year later in 2018 are presented in Table 1. These data are intended to demonstrate the year-to-year stability of groundwater chemistry at the experimental site, as no significant changes were observed over the one-year interval. As shown in Table 1, the KURT groundwater maintained a temperature of 15–16 °C and a pH of 8.2–8.5, indicating persistent slight alkalinity. The electrical conductivity (EC) values indicate the presence of dissolved ionic species, while dissolved oxygen (DO) concentrations ranged from 3.44 to 4.62 $\text{mg}\cdot\text{L}^{-1}$, corresponding to positive redox potentials (E_h). Overall, the measured parameters exhibited no substantial temporal variation.

Table 2 presents the geochemical composition of groundwater sampled from the borehole near the KURT site in 2017 and 2018. These data are included for the same purpose, namely to confirm the year-to-year stability of groundwater chemistry relevant to corrosion- and alteration-related processes. The concentrations of Cl^- and SO_4^{2-} , which may participate in corrosion processes, were below 2 $\text{mg}\cdot\text{L}^{-1}$ and 6 $\text{mg}\cdot\text{L}^{-1}$, respectively. The concentration of K^+ , which is relevant to the illitization of montmorillonite, was measured as 0.28 $\text{mg}\cdot\text{L}^{-1}$, while other metal cations were present at very low concentrations.

In addition, historical groundwater chemistry data measured at the KURT site and surrounding boreholes between 2010 and 2018 were compiled and are provided in the Supplementary Information (Table S3) to assess long-term variability of groundwater chemistry at this geological setting. These data indicate that the groundwater composition remained relatively stable over the period considered.

Table 1

Redox-related water quality of groundwater inflow and chamber effluents measured at the end of the experiments in April 2017 and one year later in July 2018 to evaluate the year-to-year stability of the internal solution chemistry (mean $\pm$ standard error, $n = 5$; chamber temperature controlled by a thermostatic bath).

Source	Sampling date	Temp (°C)	DO ($\text{mg}\cdot\text{L}^{-1}$)	E_h (mV)	EC ($\mu\text{S}\cdot\text{cm}^{-1}$)	pH
Groundwater Inflow	April 18, 2017	15 $\pm$ 0.2	3.44 $\pm$ 0.4	162.4 $\pm$ 14.2	147.3 $\pm$ 9.7	8.25 $\pm$ 0.2
	July 9, 2018	16 $\pm$ 0.5	4.62 $\pm$ 0.5	178.5 $\pm$ 13.5	144.8 $\pm$ 10.3	8.57 $\pm$ 0.3
Chamber #1 (3.1 y)	April 18, 2017	70 $\pm$ 0.1	2.25 $\pm$ 0.2	192.2 $\pm$ 15.5	155.9 $\pm$ 11.2	8.44 $\pm$ 0.3
	July 9, 2018	70 $\pm$ 0.1	2.71 $\pm$ 0.3	171.3 $\pm$ 9.1	146.3 $\pm$ 13.6	8.50 $\pm$ 0.3
Chamber #2 (6.5 y)	April 18, 2017	70 $\pm$ 0.1	2.28 $\pm$ 0.2	176.5 $\pm$ 10.1	154.6 $\pm$ 14.2	8.46 $\pm$ 0.1
	July 9, 2018	70 $\pm$ 0.1	2.44 $\pm$ 0.1	172.3 $\pm$ 7.0	169.9 $\pm$ 18.1	8.42 $\pm$ 0.1

2.3. Experimental setup

The experimental cell was designed to simulate a specific region and condition within the buffer of EBS as depicted in Fig. 1a. The objective is to investigate the long-term reactions at the canister-buffer interface—focusing primarily on metal corrosion and its subsequent effects on the bentonite—under conditions where the buffer becomes wetted by groundwater inflow and subsequently remains in a saturated state, while also being heated by spent fuel decay heat, which was simulated by maintaining the temperature at 70 °C [7,29].

The two separate blocks were prepared for assembly; one block featured a central groove identical to the dimensions of the metal specimen to ensure a tight fit. The metal coupon was placed in the groove, the blocks were joined, and this assembly was then placed between two water-permeable Ti-filters (Fig. 1b and c). This entire set was inserted into the outer rigid cell, which has perforated holes allowing the solution inflow. The rigid cell simulates the surrounding host rock, providing confinement; This cell represents the typical configuration employed for corrosion experiments at KAERI [16,30–32].

The batch experiment was designed to simulate the interactions between corrosion cell components in a stagnant groundwater. A prepared unit cell was placed in a sealed glass container and filled with KURT groundwater. This bottle was sealed to block external air and maintained at 70 °C for 2 years to simulate the decay heat from spent fuel.

The continuous flow experiment used bath-type chambers placed in one of the galleries in KURT (Fig. 2a). Five identical corrosion cells for each of the five metal types were loaded into their respective chambers, each with a volume of 37 L (Fig. 2b), and the chamber temperature was maintained at 70 °C. Groundwater (supplied from a 150 m depth) was continuously fed into the bottom and discharged from the top at a constant flow rate of approximately 10–20 $\text{mL}\cdot\text{min}^{-1}$. This condition was selected to maintain stable geochemical boundary conditions and ensure diffusion-dominated transport without inducing physical erosion. The 3.1-year specimens were retrieved from chamber#1 (March 25, 2014–April 19, 2017), and the 6.5-year specimens were retrieved from chamber#2 (October 19, 2010–April 25, 2017).

2.4. Analytic methods for the characterization

The redox-related water quality parameters, including dissolved oxygen (DO), pH, redox potential (E_h), conductivity, and temperature, were measured using either portable multiparameter meters (Orion Star A326/A329, Thermo Scientific) on collected samples or an in-situ probe (MS5, MiniSonde5, HydroLab) connected to a packer within the borehole.

The chemical composition of the groundwater was characterized in the laboratory: cations were quantified using ICP-OES (iCAP7400 duo, Thermo Scientific), and anions were analyzed using ion chromatography (IC; ICFLEX 930, Metrohm). The $[\text{HS}^-]$ content was measured separately using a dedicated measurement kit (B50-HySuL1, ITS Inc.).

Mineralogical composition and content were analyzed using XRD at the Korea Institute of Geoscience and Mineral Resources (KIGAM). The

Table 2

Geochemical characteristics of groundwater sampled from boreholes near the KURT site, measured at the end of the experiments in April 2017 and one year later in July 2018 (mean $\pm$ standard error, $n = 5$).

Element	Concentration (April 18, 2017) (mg·L ⁻¹)	(July 9, 2018) (mg·L ⁻¹)	Element	Concentration (April 18, 2017) (mg·L ⁻¹)	(July 9, 2018) (mg·L ⁻¹)
Na ⁺	36.9 $\pm$ 3.1	23.0 $\pm$ 2.0	SO ₄ ²⁻	12.1 $\pm$ 1.8	6.19 $\pm$ 1.0
K ⁺	0.58 $\pm$ 0.1	0.28 $\pm$ 0.0	NO ₃ ⁻	0.11 $\pm$ 0.0	0.10 $\pm$ 0.0
Ca ²⁺	2.4 $\pm$ 0.3	4.48 $\pm$ 0.2	F ⁻	9.0 $\pm$ 0.3	9.35 $\pm$ 0.2
Mg ²⁺	1.35 $\pm$ 0.2	0.00 $\pm$ 0.0	CO ₃ ²⁻	6.2 $\pm$ 0.9	6.0 $\pm$ 1.3
SiO ₂	18.93 $\pm$ 1.0	15.28 $\pm$ 1.0	HCO ₃ ⁻	46.6 $\pm$ 4.7	36.6 $\pm$ 3.7
Cl ⁻	3.8 $\pm$ 0.6	2.06 $\pm$ 0.4	HCO ₃ ⁻ + CO ₃ ²⁻	52.8 $\pm$ 3.1	42.6 $\pm$ 1.3

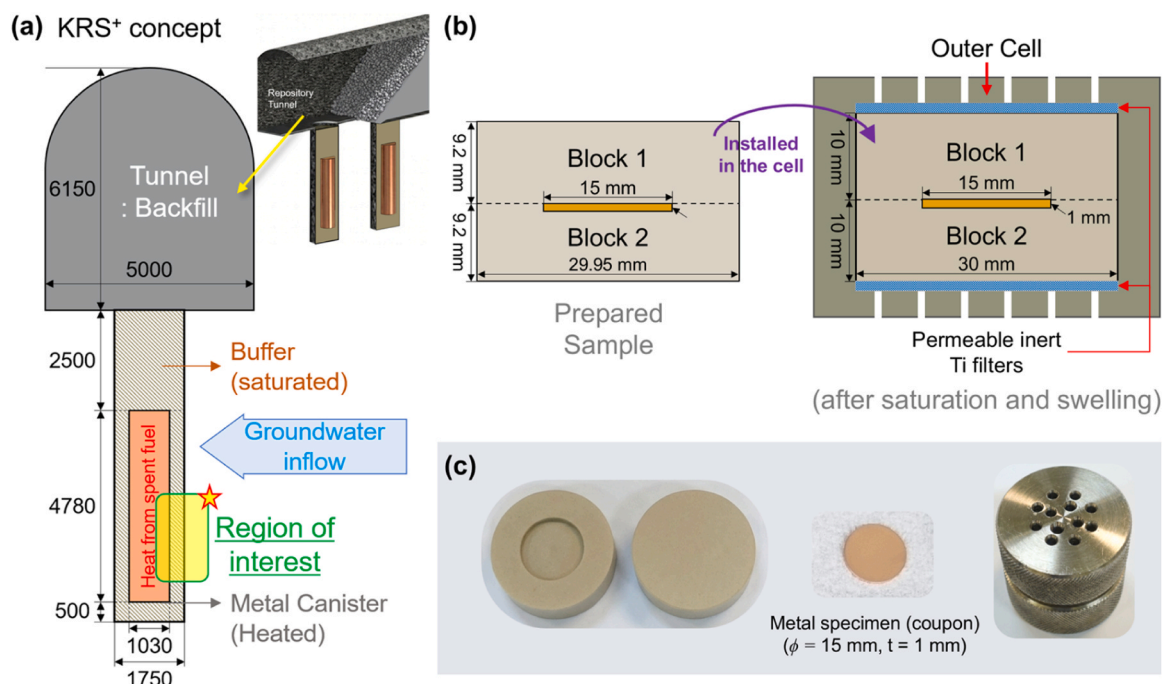


Fig. 1. (a) The KRS+ repository concept, highlighting the region of interest in this study. (b) Experimental cell configuration showing the prepared sample ($\phi = 29.95$ mm) and the cell after saturation and swelling ($\phi = 30$ mm), enclosed by a rigid outer cell and permeable inert Ti filters. (c) Photograph of the bentonite blocks, metal specimen (coupon; $\phi = 15$ mm, $t = 1$ mm), and assembled experimental cell.

XRD patterns were obtained using a Philips X'Pert-PRO MPD diffractometer with an anticathode Cu-K α radiation source ($\lambda = 1.54$ Å) operating at 40 kV and 30 mA. Samples were scanned from 3° to 65° 2 θ with a step size of 0.01° 2 θ and a scan rate of 0.04 s per step. The resulting mineral composition and quantitative content were analyzed using the SIROQUANT v4.0 program.

EDS analysis was conducted on cross-sections of the bentonite block to observe the propagation of Cu and Fe; these samples were prepared by cutting the block perpendicularly to the contact plane of the metal specimen. The analysis was conducted using a ZEISS Crossbeam 540 with a Bruker EDS system (Oberkochen, German) at Chungbuk National University. For an additional quantitative analysis of Cu and Fe propagation in the 6.5-year bentonite blocks, samples were collected by sectioning the blocks at 0.5 mm intervals from the metal-contact surface. The concentrations of Cu and Fe within these sections were then quantified via ICP-OES (I-CAP Q, Göteborg, Sweden) at the Ochang center of Korea Basic Science Institute (KBSI).

2.5. Measurement of swelling index and dry density

For the swelling index measurement, samples were physically separated into contaminated 'metal-contact' and 'non-contact' parts (Fig. 3). The retrieved block pieces were dried at 110°C and then pulverized into

a fine powder using a mortar and pestle. A precise mass of 0.6 g of this dry powder was slowly added to distilled water in a 10 mL test tube to ensure thorough wetting. After hydrating for at least 24 h, the final sedimented volume (mL) was measured and converted to the standard swelling index (mL·2 g⁻¹) for comparison.

For the measurement of block dry density, a mineral oil impregnation method was devised, as the blocks retrieved from the cells were atypical and irregular fragments; the details of the process are presented in **Supplementary Text S1** in the SI. This method calculates the dry density (ρ_d) based on the dry weight (W_d) of the target fragment, the solid particle density (ρ_s), and the void volume (V_v) determined from the mass of impregnated oil. The ρ_s of the Gyeongju bentonite was first determined to be 2.65 g·cm⁻³ by 2.12 g of bentonite using a 25 mL Gay-Lussac pycnometer, and CaCl₂ 1 M solution [33]. The porosity (n), 39.6%, was calculated using the dry density of 1.6 g·cm⁻³ and the particle density of 2.65 g·cm⁻³, according to the following relationship between ρ_d and ρ_s (Eq. (1)).

$$n = 1 - \frac{\rho_d}{\rho_s} \quad (1)$$

For the procedure, the atypical fragments were first dried at 110°C for 24 h to obtain the W_d . The dried specimens were then submerged in mineral oil ($\rho_o = 0.862$ g·cm⁻³) and placed under vacuum using an

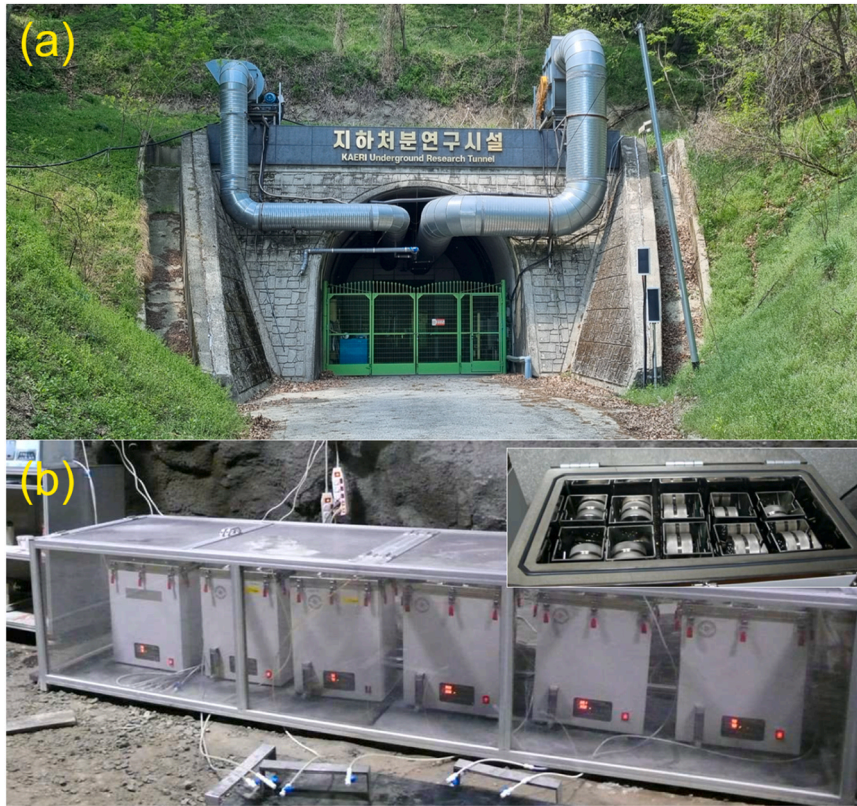


Fig. 2. (a) the KAERI Underground Research Tunnel (KURT). (b) The continuous-flow experimental setup installed in the gallery; the inset shows the assembled cells inside the chamber.



Fig. 3. Separation of bentonite block into 'metal contact' and 'non-contact' parts.

aspirator for 2 h for complete deaeration. After 3 days of immersion, the saturated specimens were removed, wiped of excess surface oil, and stabilized in air for approximately 1 day before measuring the final saturated weight (W_{sa}). The dry density was then calculated using the formula as Eq. (2):

$$\rho_d = \frac{W_d}{V_t} = \frac{W_d}{V_s + V_v} = \frac{W_d}{\frac{W_d}{\rho_s} + \frac{W_{sa} - W_d}{\rho_o}} \quad (2)$$

3. Results and discussion

3.1. Alteration of bentonite buffer under long-term groundwater exposure: batch setup

3.1.1. Mineral dissolution: pH evolution and amorphous silica precipitation

The primary observations of interest in this setup were the change in pH and the occurrence of a white, gel-like substance. In terms of pH, the initial pH of the groundwater solution within the batch reactor was 8.1, and no pH change was observed during the initial 10-day period of the experiment. Afterwards, the measured pH at the end of the 2-year experimental duration was approximately 10.9. The total amount of OH^- generated, based on a simple calculation, is approximately

1.586×10^{-4} mol ($[\text{OH}^-]$ at pH 8.1 $\ll$ $[\text{OH}^-]$ at pH 10.9).

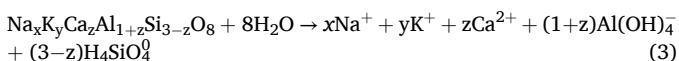
Fig. 4a and b show the solution outside the cell containing the r-Cu coupon taken at 4 months and 19 months after the experiment commenced, respectively. A white, gel-like substance was visually confirmed to be present in the exterior solution starting around the 4-month mark. By the 19-month mark, a significantly larger quantity of this seemingly identical or similar white, gel-like material had formed and was confirmed to be dispersed in the solution in a colloidal or gel form. The substance is presumed to be an amorphous SiO_2 -based aluminosilicate material, which can exist as aggregated colloids. As presented in Fig. 4c, the results of XRD analysis on the collected substance showed no distinct peaks indicative of crystalline mineral phases. Instead, it exhibited a broad band pattern around 22° (2θ), which is identical to the typical XRD pattern observed for amorphous SiO_2 materials in several previous studies [34–36].

3.1.2. Analysis of alteration mechanism and implications

Prior to the analysis, the hydration status of the bentonite block was considered. Based on empirical observations from preliminary tests, the time required for full saturation followed a diffusion-controlled trend ($\propto L^2$, corresponding to an apparent water diffusivity of approx. $8.3 \times 10^{-11} \text{ m}^2 \cdot \text{s}^{-1}$): approximately 2 weeks for a 10-mm thickness and about 60 days for a 20-mm thickness. Thus, for the 10-day experimental period, the 20-mm thick bentonite block was not fully saturated; only the outer region (several mm) was wetted. In contrast, at the 4-month and 2-year marks, the bentonite blocks were considered to be fully saturated.

At first, the absence of pH change in the solution during the initial 10 days suggests that the amount of water-soluble alkaline substances that could immediately raise the pH upon contact with water is negligible. In addition, no significant dissolution of pH-increasing minerals (e.g., feldspar, pyrite, and calcite [37–40]) appears to have occurred during this period.

This is noteworthy, as previous literature has shown that powdered feldspar in distilled water can cause a pH increase at a rate sufficiently faster than this 10-day timeframe [37]. For instance, one study reported that a 20 g sample of feldspar in 200 mL of distilled water caused a maximum pH increase of about 2.5 (from an initial pH of 7.0) after only 50 h of exposure, with the increase rate slowing between 30 and 48 h. This process generated approximately 0.315×10^{-6} mol of OH^- per g feldspar by its completion. As a general reaction mechanism, the dissolution of Na-feldspar can be expressed as shown in Eq. (3), which represents a hydrolysis reaction with water where the replacement of Na^+ by H^+ derived from water dissociation results in the release of hydroxyl ions, thereby indicating a rise in pH [38]:



To apply this to the current 10-day results, the feldspar fraction (31.2%) and dry density of $1.6 \text{ g} \cdot \text{cm}^{-3}$ were considered, along with an

estimated OH^- maximum travel distance of 4.16 mm (based on $D_a \approx 10^{-11} \text{ m}^2 \cdot \text{s}^{-1}$). If the reaction rate reported in the literature were applicable, the accessible feldspar in this region (1.47 g) should have generated approximately 0.4624×10^{-6} mol of OH^- , resulting in a final pH of around 8.55; however, no actual pH change was observed. This discrepancy strongly suggests that the feldspar-induced pH change, while rapid in powdered samples, is not a short-term effect in this compacted bentonite system, likely due to structural or diffusion limitations.

For the long-term exposure, the observed pH increase to 10.9 after 2 years was examined with respect to its potential origin from either (1) copper corrosion or (2) dissolution products of bentonite constituent minerals. Based on the observations, the pH increase is strongly suggested to be primarily driven by the dissolution of feldspar, whereas copper corrosion appears to have a negligible effect on the solution pH.

First, to evaluate the influence of copper corrosion, the corrosion phenomena expected to occur in this system were examined. After 2 years, the corrosion depth corresponding to the amount of corroded Cu was approximately $0.3 \mu\text{m}$ (Fig. S3). The potential amount of OH^- generated from this corrosion was estimated stoichiometrically, simply considering the reaction: $\text{Cu} + 0.5 \text{O}_2 + \text{H}_2\text{O} \rightarrow \text{Cu}(\text{OH})_2$. With a highly optimistic assumption, the maximum amount of OH^- corresponding to this corroded Cu amount is 1.645×10^{-5} mol, resulting in a theoretically attainable pH below 9.9; this value is 10-fold lower than the measured pH. In addition, the entire amount could not have been released in reality, given the compactness of the media. Thus, Cu corrosion could not have been the primary cause of the pH change.

In addition to the predicted low contribution of copper corrosion to the pH evolution, the white substance observed at the 2-year mark serves as supporting evidence that the dissolution of Si-based mineral induced the pH increase in the long-term exposure. This is further substantiated by the absence of initial pH change (indicating no soluble alkaline matters were present) and the mechanisms suggested by the previous studies on feldspar dissolution.

The emergence of the white gel is mechanistically linked to the observed increase in pH. SiO_2 -dissolution can be accelerated by the attack of KOH, shown as Eq. (4) [41]; for reference, quartz is known to dissolve slowly in hot alkaline solutions, and amorphous silica dissolves much faster [42–44].

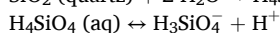
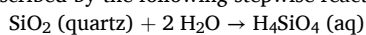


Based on these observations and established dissolution mechanisms, a sequential mechanism for the emergence of the white gel is proposed. The process begins with feldspar dissolving relatively faster, which induces an early pH increase. The resulting high pH subsequently promotes the dissolution of other SiO_2 -bearing mineral components within the bentonite, such as quartz or cristobalite (Fig. S1). The plausibility of this interpretation is further supported by the quantitative mineralogical analysis, which shows that silica-related phases (including albite, quartz, and cristobalite) collectively account for over



Fig. 4. Photographs of the exterior solution taken at (a) 4 months and (b) 19 months, showing the accumulation of a white gel-like substance. (c) XRD results of the white gel, exhibiting a broad peak characteristic of typical amorphous SiO_2 .

40 wt% of the total composition, providing a substantial source of dissolved Si species. Specifically, the dissolution of quartz and the dissociation of silicic acid under these elevated pH conditions can be described by the following stepwise reactions, Eq. (5):



Subsequently, the dissolved Si-species (Si_xO_y) re-precipitates as a white-gel in the external solution.

In summary, under the high-temperature close to 100°C and high-pressure environments that would persist in a deep geological repository, the metal canister is expected to contribute negligibly to the

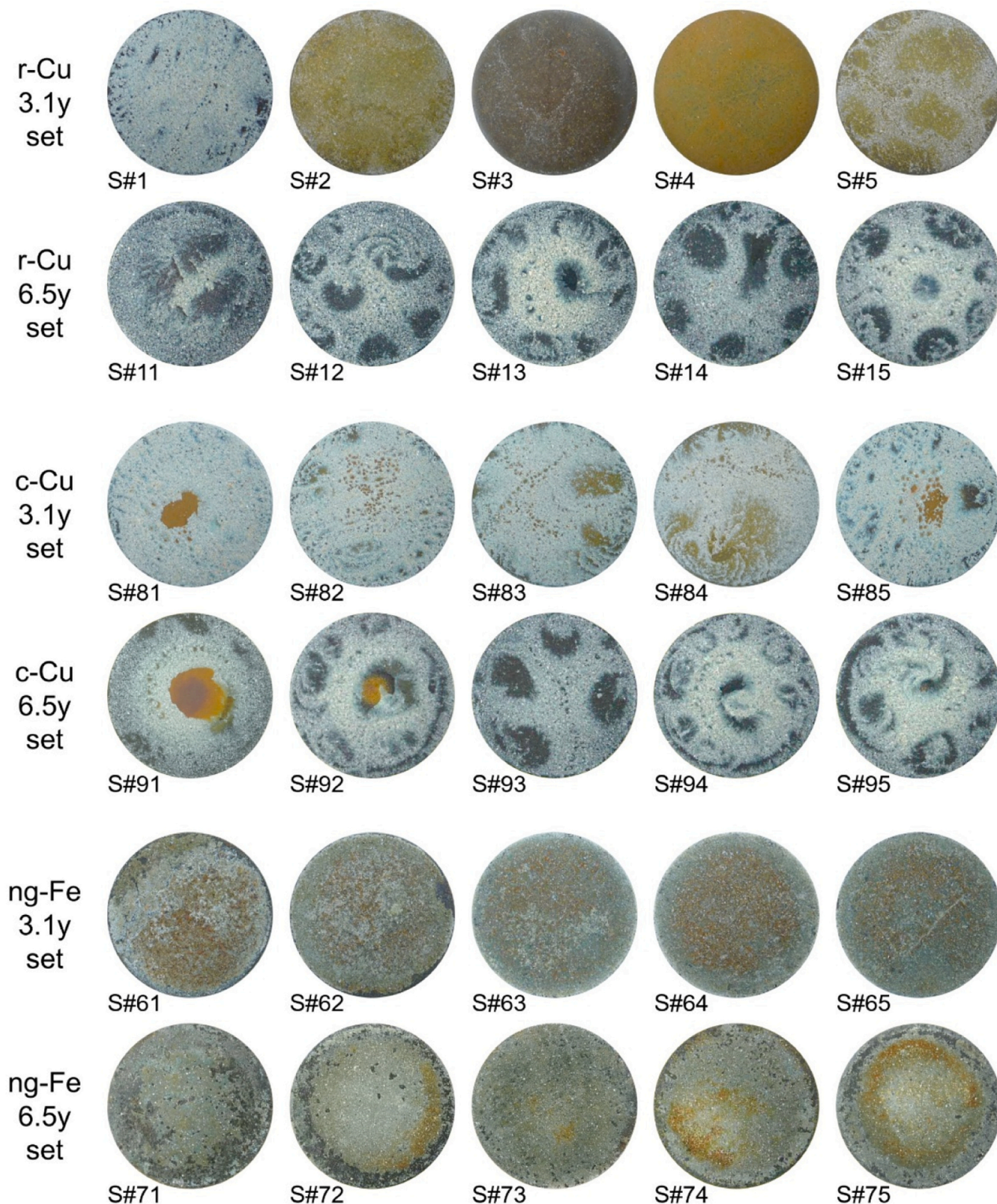


Fig. 5. Visual appearance of the retrieved r-Cu, c-Cu, and ng-Fe specimens for the 3.1-y and 6.5-y sets (note: all coupons were retrieved from individual cells; the 6.5-year samples are not the continued aging of the 3.1-year samples but are distinct specimens.).

chemical erosion of the bentonite block. However, a mechanism exists at the outermost part of the buffer, the interface between the bedrock boundary and the buffer, whereby SiO_2 -components within the bentonite can slowly dissolve along the groundwater flow. Whether the loss amount via this mechanism significantly impacts the stability of the buffer performance remains unclear, and this uncertainty presents a significant challenge for future assessment.

3.2. Alteration of bentonite buffer under long-term groundwater exposure: continuous setup

Unlike in the batch experiment, the evolution of the external solution was not investigated in this continuous flow setup. Upon retrieval, bentonite-derived reaction products were found strongly adhered to the r-Cu, c-Cu, and ng-Fe metal coupons even after ultrasonic (US) cleaning, and their metallic luster had completely disappeared. In contrast, the SS and Ti specimens retained their metallic luster after US cleaning, with no strongly adhered materials observed. Consequently, the discussion related to Ti and SS is not deeply provided here. Furthermore, this article will not cover the metal corrosion phenomena in depth, as these have been published in other literature for reference [30,32,45]. The discussion here focuses primarily on the chemical alteration of the bentonite block and its repository buffer performance concerning the two types of Cu and Fe.

3.2.1. Bentonite-derived materials on metal surfaces

The metal samples with adhered bentonite-derived materials were visually observed as shown in Fig. 5, while the uncorroded SS and Ti samples are presented in Fig. S4. The weight loss of the metal specimens, listed in Table S4 [32,45], is consistent with the observations; the loss was evaluated after removing all corrosion products with acid to expose the bare metal (Fig. S5). The SS and Ti specimens showed negligible mass loss. On the other hand, the Fe and Cu samples experienced both mass loss and strong material adherence; the mass loss was much greater for Fe.

Considering that the metal specimens and bentonite blocks were initially inserted without any chemical treatment, and noting the clear distinction in surface conditions between SS and Ti versus Cu and Fe, it is reasonable to attribute these differences, such as luster, corrosion, and adhered materials, to chemical reactions. Based on this understanding, both visual observation and XRD analysis were employed to identify the materials present on the specimen surfaces. For interpretation, the constituent minerals of the original KJ-I bentonite [16,45–47]—montmorillonite, quartz (cristobalite), clinoptilolite, and albite (Na-feldspar)—were used as baseline phases for the XRD analysis.

For the r-Cu 3.1-y set (S#1–S#5), notable visual variability was observed despite identical experimental conditions. S#1 exhibited abundant white material associated with a dark copper appearance, whereas S#4 showed predominantly yellow/red material (presumed Cu_2O) with minimal white deposits. The other samples displayed varying ratios of these white and yellow/red substances, but a correlation can be noted where a darker copper color was associated with a greater adherence of white material. For the r-Cu 6.5-y samples (S#11 to S#15), all five specimens appeared visually similar; white material was observed on the surfaces, but the yellow/red material seen in the 3.1-y set was entirely absent, and all of them had a blackish color on the base. In the XRD analysis, the 3.1-y samples primarily showed detections of Cu itself and primary bentonite components (montmorillonite, quartz, and albite). Anorthite ($\text{CaAl}_2\text{Si}_2\text{O}_8$, Ca-feldspar) was identified only in S#1, which was the only sample among the 3.1-y set to exhibit abundant white precipitates. In the S#4 sample, anorthite was not detected; instead, it showed the presence of cuprite, Cu_2O . The 6.5-y samples all primarily showed the detection of both primary bentonite components (montmorillonite, quartz, albite, and clinoptilolite) and anorthite.

For the c-Cu 3.1-y set, designated as S#81 to #85, all duplicates appeared visually similar, characterized by a high proportion of white

material on the surface. Unlike the r-Cu set, these samples exhibited distinct red/yellow patches (presumed Cu_2O) in some unadhered areas (e.g., S#81 and S#85). Additionally, even where white material was adhered, the underlying base color appeared predominantly yellow/red. For the c-Cu 6.5-y samples (S#91 to #95), the specimens were divided into two distinct visual groups. The first group (S#93, #94, and #95) showed adherence of white material with a blackish base, appearing similar to the r-Cu 6.5-y case. The second group (S#91 and #92), however, was distinct, exhibiting some reddish patches with white material on a black base. In the XRD analysis, the 3.1-y samples showed bentonite minerals as well as the presence of anorthite. For the 6.5-y samples, bentonite components (montmorillonite, quartz, albite, and clinoptilolite) were primarily detected, along with Cu_2O (identified as the red patch).

For the ng-Fe 3.1-y set (S#61–S#65), all specimens showed similar appearances, characterized by abundant white and red/greenish materials, presumed to be iron oxides. For the ng-Fe 6.5-y samples (S#71 to #75), a darker red color was observed, with less white material, and the base appeared more exposed. In the XRD analysis, the 3.1-y samples showed detections of Fe, bentonite components (montmorillonite, quartz, and clinoptilolite), anorthite, and aragonite, which was not detected in the Cu cases. The 6.5-y samples showed an almost identical composition to the 3.1-y set.

The white substances observed only on the Cu and Fe specimens suggest that Ca^{2+} was released from the Ca-bentonite by corrosion products (Cu^{2+} and Fe^{2+}) [48–50] and subsequently participated in secondary mineral formation. However, this observation alone is insufficient to conclude that Ca^{2+} directly promotes corrosion reactions. The formation of these white materials is proposed to originate from the following mechanisms: (1) Precipitates from dissolved feldspar, such as anorthite ($\text{CaAl}_2\text{Si}_2\text{O}_8$, Ca-feldspar) formed by Ca^{2+} precipitation, or reprecipitated albite (Na-feldspar); (2) Precipitates, such as aragonite (CaCO_3), formed from carbonate (originating from groundwater) reacting with Ca^{2+} ; and (3) Reprecipitation of dissolved Si-minerals.

Notably, aragonite was detected exclusively in the ng-Fe samples. Aragonite can be formed preferentially over calcite when Ca^{2+} and CO_3^{2-} react above 60°C [51], which aligns with the 70°C condition of this experiment. This selectivity for ng-Fe strongly suggests that the Ca^{2+} involved in its formation originated from the bentonite, rather than from the groundwater, since a greater release of Ca^{2+} was likely induced by the larger molar quantity of Fe^{2+} than Cu^{2+} ; although aragonite was not detected by XRD in the Cu samples, it is highly possible that it also formed, but in quantities below the detection limit.

In summary, the r-Cu, c-Cu, and ng-Fe specimen surfaces with adhered bentonite-derived materials revealed that, in addition to the primary bentonite constituent minerals, anorthite and aragonite were the phases showing meaningful differences in detection. Anorthite is a Ca-feldspar, and aragonite is a CaCO_3 polymorph; their occurrence indicates a significant involvement of Ca^{2+} in interfacial reactions at the metal–bentonite boundary. These results demonstrate that corrosion processes at the metal surface can trigger subsequent reactions with bentonite-derived materials through Ca-mediated pathways. From the perspective of repository safety, understanding both the extent of these corrosion-induced reactions and their spatial influence on the bentonite buffer remains a critical issue for long-term performance assessment.

3.2.2. Propagation of corrosion products into the bentonite block

This section aims to analyze alterations from the perspective of the bentonite block itself, whereas the previous section described the bentonite-derived materials strongly adhered to the metal surface, primarily as a result of metal corrosion. The discussion here will similarly focus on the cases of corroding specimens: r-Cu, c-Cu, and ng-Fe.

The disassembled blocks of the 6.5-y set are presented in Fig. 6. Upon visual inspection, a distinct darkening was observed in the region of the bentonite block in contact with the metal, compared to the non-contact surfaces, in all cases where corrosion occurred; this discoloration is

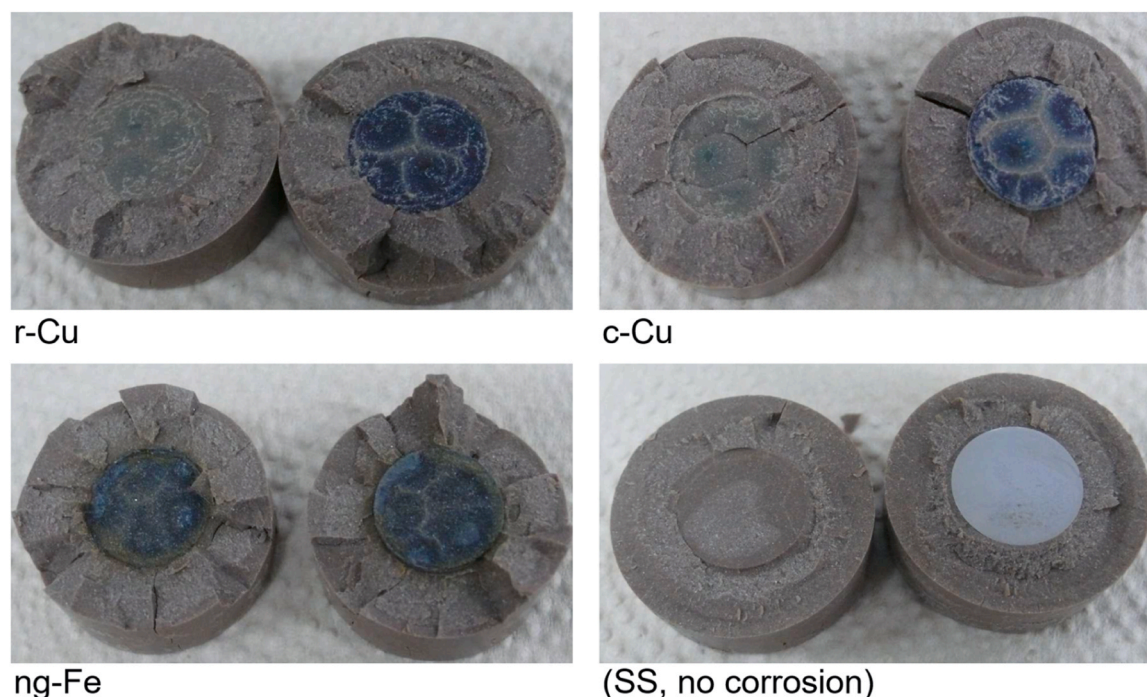


Fig. 6. Disassembled bentonite blocks (6.5-y sets) showing metal-contact surfaces; distinct discoloration is observed for r-Cu, c-Cu, and ng-Fe compared to the clean SS case.

attributed to the interaction with the metal specimen. Cross-sections of the blocks were examined to investigate propagation perpendicular to the contact interface (Fig. S6). The penetration in Cu samples was too thin to be distinguished by the naked eye, while ng-Fe displayed a distinct 2–3 mm dark layer; deeper penetration was observed in more severely corroded sample.

The EDS analysis was performed on the cross-section to quantitatively analyze the perpendicular propagation from the metal specimen contact plane (Table 3, Fig. 7, and Fig. S7); directional arrows have been included in Fig. 7 to visualize the approximate penetration trends rather than precise quantitative profiles. The results for r-Cu and c-Cu are described first for comparison of Cu cases, followed by the data for ng-Fe. For r-Cu, the results for the 3.1-y (Figs. 7a) and 6.5-y (Fig. 7b) sets both showed a clear concentration gradient moving away from the contact interface. The distinguishable gradient of the Cu signal extended to approximately 0.3 mm at 3.1 years and 0.35 mm at 6.5 years. In the

case of c-Cu, the 3.1-y sample (Fig. 7c) showed a concentration gradient that was less distinct compared to r-Cu, extending to a maximum of about 0.2–0.25 mm. By 6.5 years (Fig. 7d), the signal extended to approximately 0.3–0.35 mm. Beyond these respective depths, the signal returned to the baseline corresponding to the original Cu content of the bentonite. In addition, ICP-OES analysis, which offers higher resolution than EDS, was conducted on the 6.5-y r-Cu and c-Cu bentonite samples, collected at 0.5 mm intervals (Fig. S8) (This analysis was not performed for the 3.1-y samples, as the expected migration range was less than 0.5 mm). The ICP-OES results indicate that the migration distances of r-Cu and c-Cu are nearly identical, with Cu detected up to approximately 2.0 mm for both 6.5-y samples.

For ng-Fe, the EDS results showed that the Fe signal extended to approximately 2 mm at 3.1 years (Figs. 7e and 7f). For the 6.5-y specimen, quantitative EDS analysis was used instead (Fig. 7g; due to measurement error and signal interference). A comparison of the near-contact regions showed a higher Fe-K α 1 signal intensity for the 6.5-y case (1.02%) than for the 3.1-y case (0.93%). For the 6.5-y sample, ICP-OES analysis was also performed according to the distance (Fig. S8), and the results showed Fe penetration detected up to a 2.0–2.5 mm range.

Table 3

Averaged elemental compositions (atomic %) of the bentonite block cross-sections from line-EDS analyses for the 3.1- and 6.5-year r-Cu, c-Cu, and ng-Fe cases; near-contact area is 0–0.9 mm from the metal specimen surface; non-contact area is 0–0.9 mm from the outer surface.

Cases	Element	Near contact area (3.1-y)	Near contact area (6.5-y)	Non-contact area (6.5-y)
		(Atomic %)	(Atomic %)	(Atomic %)
r-Cu	Cu-K	0.07	0.12	0.01
	Na-K	0.72	0.49	0.81
	Ca-K	0.75	0.66	0.68
	Si-K	16.27	13.16	14.33
c-Cu	Cu-K	0.01	0.38	0.01
	Na-K	0.89	0.32	0.49
	Ca-K	0.68	0.63	0.61
	Si-K	16.78	14.42	13.98
ng-Fe	Fe-K	0.93	1.02	0.63
	Na-K	0.79	0.78	0.58
	Ca-K	0.84	0.72	0.77
	Si-K	15.39	14.42	13.61

3.2.3. Proposed governing mechanism

It should be noted that this section does not aim to provide an exhaustive or specialized analysis of metal corrosion mechanisms; rather, it serves as an auxiliary description to support the understanding of how these processes facilitate the subsequent chemical and mineralogical alteration of the bentonite system. To speculate on the mechanism by which corrosion affects bentonite, the potential corrosion products under this environment were first considered. The potential primary corrosion products of Cu include dissolved CuCl₂, precipitated Cu₂O, dissolved Cu(II), precipitated CuCl₂·3Cu(OH)₂, and precipitated Cu₂S or CuS [13,15,18–21]; CuO(II)(s) is not directly generated from Cu; it is formed by the oxidation of Cu₂O and is therefore considered a secondary corrosion product. Given the low Cl[−] ($\approx 5 \times 10^{-5}$ M) and aerobic conditions ($E_h = 170$ mV, and 72 mV for each chamber), the primary Cu corrosion products considered were precipitated Cu₂O and dissolved Cu

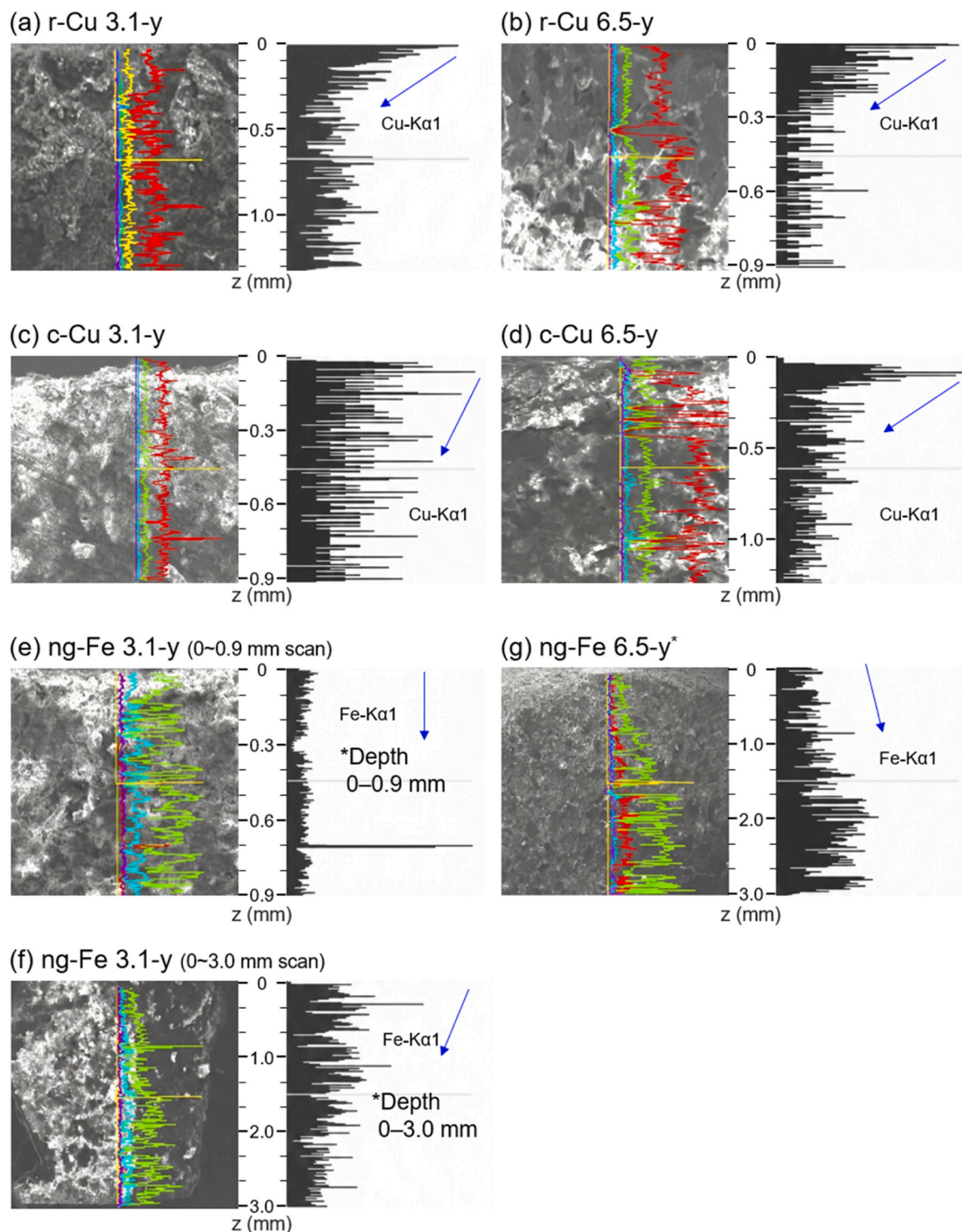


Fig. 7. Cross-sectional EDS line-scan profiles showing metal migration perpendicular to the metal-bentonite interface. (a–d) Cu-K α 1 intensity profiles for the r-Cu and c-Cu specimens; (e–g) Fe-K α 1 intensity profiles for the ng-Fe specimens after 3.1 years measured over scan lengths of (e) 0.9 mm and (f) 3.0 mm; and (g) Fe-K α 1 intensity profile for the ng-Fe specimen after 6.5 years. The colored lines represent the concentration profiles of the analyzed elements, and the arrows indicate the approximate penetration trends (see detailed legends in Fig. S7). The position $z = 0$ corresponds to the metal contact interface, while $z > 0$ indicates the propagation distance. *The ng-Fe 6.5-y profile was obtained from a specimen measured in a tilted orientation.

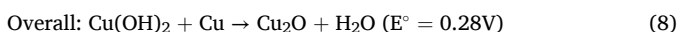
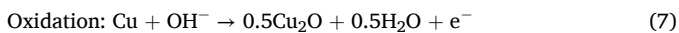
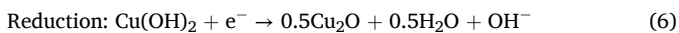
(II) [52–54].

In a system governed by this alkaline mechanism with sufficient oxygen, as confirmed by the batch results and groundwater data, $O_2 + 2 H_2O + 4e^- \rightarrow 4OH^-$ can act as the cathodic reaction, while the oxidation of the substrate Cu serves as the electron source [54]. Regarding the source of oxygen in corrosion studies, while groundwater

infiltration certainly supplies a portion of the oxidant, it is a common consensus that the air unavoidably trapped during the assembly process accounts for a very large share. If this initial oxygen starts off dissolved in the groundwater upon saturation, the corrosion process is not simply a matter of oxygen diffusing through the compacted clay from an external source; instead, oxygen available for contact is present from the

initial stage. It is generally agreed that this trapped amount is sufficient to drive significant early-stage corrosion [55].

Once saturation occurs and the solution penetrates the bentonite matrix, this initial oxygen becomes aqueous oxygen and immediately participates in the redox reaction. Following this, the dissolved Cu ions (predominantly Cu^{2+}) approach saturation in the limited water volume, precipitating as $\text{Cu}(\text{OH})_2$ in a colloidal or gelatinous form. The formation of Cu_2O then occurs via the subsequent reduction of this $\text{Cu}(\text{OH})_2$ through galvanic coupling with the substrate Cu. This thermodynamically favorable overall reaction ($E^\circ \approx 0.28 \text{ V}$), which couples the reduction of $\text{Cu}(\text{OH})_2$ with the oxidation of the Cu substrate, as following comproportionation reaction, Eqs. (6)–(8) [54]:



This Cu_2O forms first on the Cu substrate and can be further oxidized to CuO by dissolved O_2 . A schematic illustration of the overall system discussed herein is proposed in Fig. 8.

The visual observation that the 3.1-y samples were predominantly red (Cu_2O), while the 6.5-y samples were mostly dark (CuO) is consistent with a structure of the inner Cu_2O and outer CuO duplex layer described for alkaline-based mechanisms. Thus, it is highly probable that the corrosion formed via a mechanism involving $\text{Cu}(\text{OH})_2$, and the primary Cu corrosion products were Cu^{2+} and $\text{Cu}(\text{OH})_2$. This is also consistent with the behavior of Cu^{2+} in Ca-bentonite speculated above, therefore, that in this system, Cu^{2+} migrated into the bentonite block after corrosion occurred, leading to an interaction with the bentonite [50]. Similarly, considering the pH, the migrating species for iron is expected to be primarily Fe^{2+} , although the detailed mechanisms for Fe are not discussed further.

One notable point is the apparent inconsistency between the corrosion amount and the propagation distance observed for the r-Cu and c-Cu samples in the 3.1-y sets. The corrosion amount of c-Cu was approximately 1.65 times greater than that of r-Cu ($2.819 \mu\text{m}$ vs. $1.703 \mu\text{m}$); however, their propagation distances were nearly identical. Furthermore, the increase in corrosion amount during the 3.1–6.5-year period was considerably similar for both (with respective increases of $1.263 \mu\text{m}$ for r-Cu and $1.091 \mu\text{m}$ for c-Cu), and their propagation distances at 6.5 years also remained similar.

Based on these results, the following sequential mechanism is proposed: Within the 3.1-year period, the passivation of r-Cu was completed first, followed by the completion of passivation for c-Cu. Until full passivation was achieved, insoluble corrosion products continued to

form [55]. For each r-Cu and c-Cu, after passivation was individually completed, mobile Cu was generated at a relatively slow rate from these different starting points until the 6.5-year mark, and this reaction likely proceeded via a nearly identical mechanism for both Cu specimens; this process can be understood as being analogous to the formation of the inner Cu_2O and outer CuO duplex layer under alkaline conditions, as described in other literature [51,53,55].

In the 3.1-y sample comparison, the inconsistency suggests that c-Cu, compared to r-Cu, formed a relatively higher proportion of solid-phase corrosion products rather than dissolved Cu ions during the same period. A possible explanation is that the c-Cu possessed higher purity, which allowed passivation to proceed to a greater extent and resulted in the formation of more insoluble material. Unfortunately, specific purity information for the two Cu types was not secured at the time of the experiment. However, a general explanation can be proposed: the cold-spray method—a low-temperature process that induces negligible oxidation during manufacturing [56]—allows c-Cu to achieve higher purity than r-Cu, assuming the raw powder for c-Cu was purer than the raw material used for r-Cu.

3.2.4. Order-of-magnitude discussion of migration phenomena

The discussion in this section primarily focuses on the migration behavior of Cu^{2+} in compacted Ca-bentonite. It should be emphasized at the outset that the following analysis is intended as an order-of-magnitude evaluation, aimed at a rough comparison with values reported in previous studies, rather than a rigorous determination of transport parameters.

Under the assumption that diffusive transport governs the migration process, the concentration profile in a semi-infinite medium is described by Fick's second law (Eq. (9)).

$$\frac{\partial C}{\partial t} = D_a \frac{\partial^2 C}{\partial x^2} \quad (9)$$

In diffusion-controlled systems, a practical characteristic migration distance x is often defined as the location where the concentration decreases to $\text{erfc}(1)$, i.e., approximately 15.7% of the initial surface concentration ($C/C_0 = 0.157$) [15,57]. In a diffusion-dominated regime, an order-of-magnitude scaling yields the following relations, where x and t denote the characteristic migration distance and exposure time, respectively (Eq. (10)).

$$\begin{aligned} \frac{\partial^2 C}{\partial x^2} &\sim \frac{C}{x^2} \\ \frac{\partial C}{\partial t} &\sim \frac{C}{t} \end{aligned} \quad (10)$$

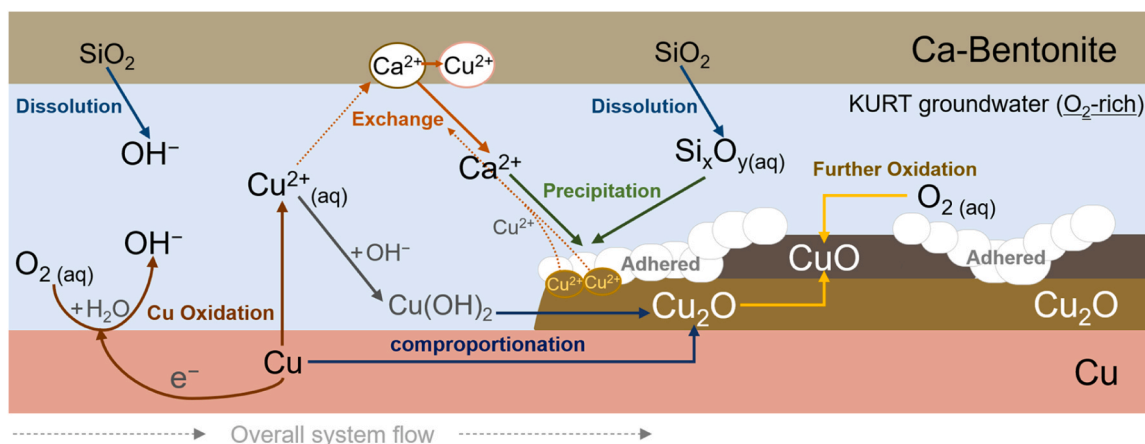


Fig. 8. Proposed overall reaction system including copper corrosion, cation exchange, and white-gel formation (precipitation) at the Ca-bentonite buffer interface with O_2 -rich groundwater.

Substituting these relations into Eq. (10) gives

$$\frac{C}{t} \sim D_a \frac{C}{x^2}; D_a \sim \frac{x^2}{t} \quad (11)$$

For a rough calculation, by substituting an exposure time of 6.5 years and a diffusion migration distance of 2 mm—evaluated from the ICP-OES results discussed above (Fig. S8)—into Eq. (11), the apparent diffusivity D_a is estimated to be approximately $4.85 \times 10^{-15} \text{ m}^2 \cdot \text{s}^{-1}$. In porous media, the relationship between the apparent diffusivity (D_a) and the pore diffusivity (D_p) can be expressed as shown in Eq. (12) [15].

$$D_a = \frac{D_p}{1 + K_d \rho_d \epsilon} \quad (12)$$

A POSIVA report stated a calculated D_a value of $2.9 \times 10^{-13} \text{ m}^2 \cdot \text{s}^{-1}$, and a corresponding experimental value of $4.0 \times 10^{-13} \text{ m}^2 \cdot \text{s}^{-1}$ at 50°C . Their calculation for the apparent Cu(II) diffusivity was based on the Eq. (12) above, where D_p is the pore diffusivity ($10^{-10} \text{ m}^2 \cdot \text{s}^{-1}$), K_d is the distribution coefficient ($0.1 \text{ m}^3 \cdot \text{kg}^{-1}$), ρ_d is the dry density ($1350 \text{ kg} \cdot \text{m}^{-3}$), and ϵ is the buffer porosity (0.44) [15].

When comparing the values reported in the literature with those obtained in the present study, the difference between the two calculated D_a values (10^{-13} versus $10^{-15} \text{ m}^2 \cdot \text{s}^{-1}$) can be attributed to differences in the parameter sets employed. The report applied a K_d value of $0.1 \text{ m}^3 \cdot \text{kg}^{-1}$ (representative of MX-80, Na-bentonite), whereas the K_d for Ca-bentonite is much larger, reported to be on the scale of $2.0 \text{ m}^3 \cdot \text{kg}^{-1}$ [58,59]. Furthermore, the adopted pore diffusivity D_p value of $10^{-10} \text{ m}^2 \cdot \text{s}^{-1}$ corresponds to Na-bentonite with a dry density of $1.35 \text{ g} \cdot \text{cm}^{-3}$. However, according to reported diffusion coefficient and sorption data [60], for Ca-bentonite compacted to a higher dry density of $1.6 \text{ g} \cdot \text{cm}^{-3}$, a lower D_p value on the order of $10^{-10} \text{ m}^2 \cdot \text{s}^{-1}$ can be selected. Considering the plausible ranges of K_d and D_p , the D_a value obtained in this study appears realistic, and its difference from the value reported by POSIVA can be explained by the differences in the adopted conditions.

A significant factor contributing to the D_a discrepancy may be the aging atomic diffusion of Cu into bentonite, a phenomenon not considered in the previous report, which is characterized by the change over time in the extractable Cu fraction of exogenous Cu in Ca-bentonite, as measured via a sequential extraction method [27]. Over an incubation duration of 370 days, exchangeable Cu decreased from 18.0% to 1.1%, and EDTA-extractable Cu dropped from 70.3% to 22.5%. The portion of 0.5 M HCl-extractable Cu, representing very strongly bound Cu, rose from 0% to 37% within 30 days, while the 6 M HCl and residual Cu fractions showed a continuous increase throughout the 370-day period. The author interpreted these results as Cu^{2+} undergoing atomic diffusion into the interlayer or crystalline interior, thereby becoming less extractable. This aging phenomenon has also been reported by another author in a 500-day experiment [28]. Considering the 6.5-year experimental period of this study, it is reasonable to conclude that this aging phenomenon contributed to the observed results, in addition to the diffusive movement described by the pore apparent diffusivity.

The ng-Fe case showed that Fe had approximately a 10-fold greater migration distance compared to the Cu cases; therefore, the D_a is expected to differ by two orders of magnitude. The origin of this D_a discrepancy can be understood by examining the differences in K_d and D_p between Fe and Cu. In the case of K_d , the sorption of Fe^{2+} into bentonite is known to be preferable to that of Cu^{2+} [61]. However, Fe^{2+} showed a much longer migration distance than Cu^{2+} , which is a contradictory result, suggesting that the contribution of behavior related to D_p must be much higher.

This is supported by calculations indicating that the amount of Fe^{2+} generated by corrosion likely exceeded the CEC of the bentonite, leading to the migration of excess Fe as an aqueous phase. The CEC value of KJ-I bentonite is $0.62 \text{ meq} \cdot \text{g}^{-1}$, equivalent to $3.1 \times 10^{-4} \text{ mol} \cdot \text{g}^{-1}$ for $2+$ ion. Given a block dry density of $1600 \text{ kg} \cdot \text{m}^{-3}$, the corresponding CEC is calculated to be $8.765 \times 10^{-5} \text{ mol per mm of penetration depth}$. The

total corrosion quantity of 6.5-year ng-Fe was $7.489 \times 10^{-4} \text{ mol}$ (3.5 times more than $2.144 \times 10^{-4} \text{ mol}$ observed for c-Cu). This total Fe amount corresponds to approximately 4.27 times the total CEC of the 2 mm region, confirming that the corrosion products significantly exceeded the available CEC.

Furthermore, comparing the K_{sp} of Cu and Fe hydroxides under the pH of this study, that of Fe has a larger value by a 10^4 scale, meaning Fe^{2+} ionization is more favorable than Cu^{2+} . Therefore, the portion of Fe^{2+} that migrated without being sorbed (after exceeding the CEC) was much greater, which would be the main reason for the difference in travel distance.

3.3. Investigation of mineral and montmorillonite alteration

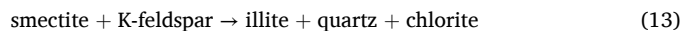
Given that bentonite is a clay material, its alteration should be understood in terms of mineralogical changes. Among its constituents, montmorillonite is directly linked to the safety performance of the repository, with a provisional content recommendation of approximately 70% for buffer materials [11–13]. The KJ-I bentonite used in this study, despite its slightly lower montmorillonite content, was selected based on its availability as a domestically supplied material with the highest achievable quality; such prioritization of domestic availability has been adopted as a bentonite selection criterion in several countries, including Japan and the Czech Republic.

Table 4 presents the XRD quantitative analysis results for the original bentonite, the 2-year batch sample, and the specimens from the 3.1- and 6.5-year continuous experiments, including both non-contact and ng-Fe contact cases. Compared to the original bentonite, albite and quartz content were decreased in all exposed samples. In contrast, calcite contents increased in all samples, and the contents of cristobalite and montmorillonite also increased slightly.

The decrease in quartz and albite appears consistent with the dissolution mechanisms observed in the batch experiment. The increase in cristobalite fraction is likely due to the dissolution loss of other minerals; the cristobalite formation from amorphous silica requires the temperature over 800°C , and is thus unfeasible [62]. The increase in calcite would be attributed to the reaction of Ca^{2+} from bentonite with CO_3^{2-} in groundwater, an inference which aligns with the earlier discussion involving the exclusive presence of CaCO_3 on the larger corroded ng-Fe samples.

For the montmorillonite alteration, SKB has detailed the transformation processes and their corresponding analogues to explain the stability of montmorillonite by reflecting underground conditions [63]. The most likely and exclusive scenario for montmorillonite degradation would be the transformation to illite, which proceeds through SiO_2 -dissolution accompanied by quartz precipitation, layer charge increase due to Si-substitution, and the possible fixation of potassium.

In the present study, illitization was not unambiguously identified from the XRD results, as the characteristic illite reflection ($10 \text{ \AA}$ d-spacing near $2\theta \approx 8.8^\circ$) was indistinguishable from the background signal. However, the potential for the montmorillonite-to-illite reaction was evaluated based on solution chemistry parameters, including K^+ concentration, Si activity, pH, and temperature. The critical conditions for illitization are generally reported as Si activity condition of $\log \{[\text{H}_4\text{SiO}_4]\} < -3$, a potassium condition of $\log \{[\text{K}^+]/[\text{H}^+]\} > 2$, and temperatures above 40°C [63]. Assuming pH 8.5 and unit activity, the threshold K^+ concentration is approximately $0.0124 \text{ mg} \cdot \text{L}^{-1}$. The general reaction pathway for illitization is expressed as Eq. (13) [64]:



Considering the experimental temperature of 70°C , the measured K^+ concentration ($0.28\text{--}0.38 \text{ mg} \cdot \text{L}^{-1}$) significantly exceeding the threshold, and the observed feldspar dissolution, the conditions are mechanistically favorable for illitization. Therefore, under these environmental conditions, it is likely that illitization occurred, although the

Table 4

Comparison of mineral compositions for the original source, batch, and continuous experimental samples.

Minerals (XRD)	Source		Continuous			
	Original	Batch				
	KJ-I source	2.0-y	3.1-y no contact	3.1-y ng-Fe contact	6.5-y no contact	6.5-y ng-Fe contact
Montmorillonite	57	63.8	61.6	59.7	57.7	61.2
Albite	31.2	26.5	25.7	28.3	30.4	25.7
Quartz	7.1	4.5	5.5	5.8	5.8	6
Cristobalite	2.9	3.3	4.6	3.7	4	4
Calcite	1.8	1.9	2.5	2.5	2.1	3.1

transformation was insufficient to be detected by XRD.

The slight increase in montmorillonite content observed via XRD is likely attributable to technical uncertainty or a relative proportion increase resulting from the dissolution of other minerals, rather than a definitive net growth. However, from a mechanistic perspective, the transformation pathway from albite to montmorillonite is plausible at 70°C, as similar transformations have been reported even at 47°C [65, 66]. Therefore, while no direct evidence was obtained in this study, the potential formation of montmorillonite from albite dissolution over the 6.5-year duration cannot be entirely excluded.

Although the montmorillonite structure showed almost no difference by the metal contact condition in this experiment, an additional possibility must be considered: a report on the phenomenon of Cu becoming strongly bound to the bentonite structure by an atomic diffusion mechanism via aging [27]. This process allows Cu to participate in the atomic structure, existing within the structure and exerting a semi-permanent effect, with Cu remaining as a residual component over 300–500 days. This mechanism has not been separately considered in previous studies, and research on its potential effects on the internal structure of montmorillonite should be investigated in the future.

3.4. Buffer performance

To ensure the effective isolation of hazardous materials underpinned by the mineralogical stability discussed above, the buffer must maintain

adequate macroscopic performance. Swelling properties are essential for physical flow retardation by sealing gaps to ensure a diffusion-dominated transport regime, and the dry density serves as a comprehensive metric that represents both the physical integrity and the chemical abundance of sorption sites.

Consequently, the phenomena observed in this study, including the leaching of constituent minerals, the formation of precipitates, the emergence of ionic corrosion products (Cu^{2+} and Fe^{2+}), and conditions favorable for mineral alteration, must be interpreted as potential drivers for the degradation of this retardation capacity. Since a decline in these values can be judged as a direct reduction in the HLW containment capability, variations in these performance metrics, however slight, observed through long-term monitoring must be treated with critical importance and warrant further in-depth investigation.

3.4.1. Evaluation of changes in swelling properties

The ultimate goal of assessing the alteration of the bentonite block upon environmental evolution is to explore the rate and extent of degradation in buffer performance; this section evaluates the swelling index (defined as the volume of 2 g of fully hydrated bentonite, per ASTM D5890), one of the key performance metrics. The ng-Fe samples, which exhibited the most significant corrosion, were investigated, and the results are presented in Fig. 9. Samples were designated based on duration and contact status ('c' = contact, 'n' = non-contact) as 3.1-y,c/n, and 6.5 y,c/n. Due to the limited quantity of the interface material, the

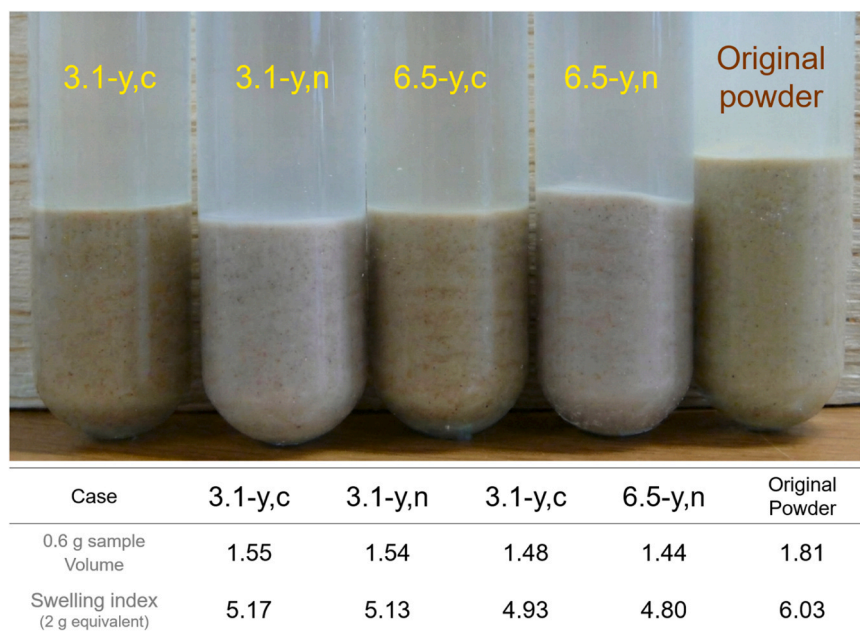


Fig. 9. Swelling characteristics of bentonite samples from continuous ng-Fe experiments (3.1 and 6.5 years), comparing contact and non-contact interfaces with the original powder. Note: Sample heights are not directly comparable; the figure is intended to illustrate the relative reduction in swelling compared to the original powder.

swelling volume was measured using 0.6 g of powder and converted to the standard index ($\text{mL} \cdot 2 \text{ g}^{-1}$) by multiplying the measured volume by a factor of 2.0/0.6. As shown in Fig. 9, all bentonite samples exposed to groundwater showed a slightly lower swelling index compared to the original bentonite. Furthermore, there appears to be a decreasing trend in the swelling index with exposure time. Notably, the quantitative data reveals that the non-contact parts had slightly lower swelling indices than the metal-contact parts for the same duration.

It appears that Ca-bentonite did not experience significant changes in swelling due to 2+ -valence ion corrosion products. Although systematic analyses clearly presenting experimental results of swelling degradation for the substitution of Ca-bentonite with 2+ -cations are unavailable, an indirect assessment supports this observation. A report comparing the swelling pressure in compacted blocks showed values of 7.5–15 MPa for Na-bentonite, 5.2 MPa for Fe-bentonite, and 5.2 MPa for Ca-bentonite [67]. This comparison implies that a meaningful decrease in the swelling capacity of Ca-bentonite due to simple substitution by 2+ -valence cations is not expected.

It should be noted that a pathway exists where Fe, when incorporated into the montmorillonite structure, transforms into berthierine, characterized by a non-swelling d -spacing of 7 Å [67]. While direct evaluation of d -spacing was not feasible as the expected amount of structural change is considered to be below the detection limit of XRD, the macroscopic results provide a basis for logical inference. Since the swelling index observed in this study showed no significant difference between the contact and non-contact parts, it can be deduced that the fundamental interlayer structure was not transformed by Fe. Therefore, the discoloration is primarily attributed to pore-filling Fe-oxide precipitates, although the possibility of minor Fe-induced alteration cannot be entirely excluded.

The decreasing trend over time and the lower swelling index for the non-contact part compared to the metal-contact part implies that the degradation in swelling capacity is mainly due to long-term exposure to groundwater. The illitization mechanism can be considered a cause of this degradation, as the swelling capacity of illite is only about 20% that of montmorillonite [68], a possibility noted previously.

3.4.2. Evaluation of changes in bentonite block density

The most intuitive indicator for evaluating the performance of a bentonite buffer in a repository is the dry density, as its correlation with key performance metrics has been established [8–12]. To assess the changes in dry density and porosity, the blocks of ng-Fe and SS samples—representing the most and least corroded cases, respectively—were separated into multiple fragments, and three individual parts were measured for each case ($n = 3$) as shown in Table 5.

The results indicated no drastic time-dependent trend between 3.1 and 6.5 years of exposure. Although the dry density in all cases decreased compared to the initial value ($1.730 \text{ g} \cdot \text{cm}^{-3}$), the samples exposed for 6.5 years exhibited slightly higher dry densities than those

Table 5

Comparison of dry density and porosity for the original block and experimental samples (SS and ng-Fe) after 3.1 and 6.5 years; values are mean $\pm$ standard deviation, $n = 3$.

BlockSample	Drydensity ($\text{g} \cdot \text{cm}^{-3}$)	Porosityn (%)	Note
Original	1.730 ± 0.013	34.7 ± 0.3	Original dry density refers to the value measured immediately after block compaction.
SS 3.1-y	1.673 ± 0.012	36.9 ± 0.3	4.32 g of block; mineral oil method, measured after 1 day
ng-Fe 3.1-y	1.717 ± 0.009	35.2 ± 0.4	4.24 g of block; identical method
SS 6.5-y	1.687 ± 0.015	36.3 ± 0.1	5.22 g of block; identical method
ng-Fe 6.5-y	1.721 ± 0.014	35.1 ± 0.5	5.01 g of block; identical method

exposed for 3.1 years, a difference that remained observable even when considering the measurement uncertainty. Regarding the material comparison, the ng-Fe cases consistently showed slightly higher values than the SS cases at both time points. While the net decrease in dry density across all samples implies that mineral leaching occurred—consistent with the batch setup observations—the corroding ng-Fe samples (1.717 ± 0.009 and $1.721 \pm 0.014 \text{ g} \cdot \text{cm}^{-3}$) maintained a higher dry density than the SS samples 1.673 ± 0.012 and $1.687 \pm 0.015 \text{ g} \cdot \text{cm}^{-3}$). This suggests that the retention of Fe corrosion products, or potentially Ca-precipitates induced by the corrosion process, partially compensated for the density loss within the bentonite matrix.

According to a POSIVA report, Fe ions can form precipitates in the porous microstructure, such as metastable phases ($\text{Fe}(\text{OH})_2$, green rusts) or typical end products (magnetite, Fe_3O_4) [67]; the dark greenish-black layer observed visually in this study appears to be these Fe-oxide precipitates. However, the impact of this density increase—resulting from the retention of non-montmorillonite materials—on the overall buffer performance requires further investigation.

Although these results confirm that mineral leaching by groundwater occurs, the significance of this leaching portion or degree cannot be evaluated relative to the full-scale, considering the scale and configuration. Moreover, the nearly identical dry densities of the 3.1- and 6.5-y samples suggest that the mineral leaching does not occur continuously; rather, its speed slows down as a certain equilibrium condition is approached. This phenomenon seems analogous to the behavior observed for 7 types of feldspar, where the leaching rate also slows and reaches a plateau [37]. Here, further research is deemed necessary on the kinetics of this leaching phenomenon and its impact on buffer performance.

4. Conclusion

This study designed the configurations to simulate the near bentonite-metal contact region during the early heated period of the safety case of the KRS, aiming to observe the long-term behavior of the buffer. Through a 2-year batch and a continuous flow experiment (3.1, and 6.5 years) using real groundwater at 70°C, the following conclusions were drawn:

- 1) The formation of OH^- and amorphous silica gel in the solution is strongly presumed to be driven by a groundwater-induced dissolution sequence of intrinsic minerals, starting from feldspar dissolution, rather than by metal corrosion.
- 2) Strongly adhered precipitates, such as anorthite and aragonite, were exclusively observed on corroded Cu and Fe. The variation in corrosion amount by metal type suggests that Ca^{2+} released from the bentonite actively participated in these interfacial reactions.
- 3) Cu migration reached ~ 2.0 mm, while Fe extended to 2.5–3.0 mm after 6.5 years. The inconsistency between corrosion amount and penetration depth for r-Cu versus c-Cu demonstrates a phased generation of solid/aqueous corrosion products depending on Cu purity.
- 4) Distinct illitization of montmorillonite was not detected; however, micro-scale illitization should not be ruled out given the favorable conditions.
- 5) In terms of buffer performance, slight decreases in the swelling index and dry density were observed, primarily attributed to groundwater-induced mineral leaching rather than corrosion; notably, the accumulation of corrosion products partially compensated for the density loss.

This study is an intermediate stage of the ongoing long-term observation at KAERI. Current findings indicate that groundwater conditions exert a more dominant influence on alteration than corrosion. However, given the vast spatiotemporal scales of a repository, these 6.5-year results require cautious interpretation. Observed minor variations in buffer performance should not be disregarded over safety case

timescales; therefore, continued long-term monitoring is essential for definitive safety conclusions.

Environmental implication

High-level radioactive waste poses severe environmental threats due to radionuclide toxicity, and the deep geological repository is a globally shared disposal concept. By utilizing real groundwater at 70°C for 6.5 years, this study faithfully simulates actual repository conditions to provide experimental evidence regarding the potential migration fate of HLW. We demonstrate, with empirical and mechanistic proofs, that the buffer's containment stability is primarily affected by groundwater interactions rather than metal corrosion. These findings provide a critical geochemical basis for predicting the long-term isolation of hazardous nuclear materials, ensuring they do not migrate into the biosphere and contaminate the environment.

CRedit authorship contribution statement

Jin-seop Kim: Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition. **Jae-Deuk Kim:** Writing – review & editing, Visualization, Investigation. **Minsoo Lee:** Writing – review & editing, Validation, Supervision, Investigation, Formal analysis, Conceptualization. **Taehyung Park:** Writing – review & editing, Visualization, Validation, Resources, Investigation, Formal analysis. **Yonghyeon Lee:** Writing – original draft, Visualization, Validation, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author(s) used Gemini-3.0 Pro in order to improve grammar and readability. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.jhazmat.2026.141819](https://doi.org/10.1016/j.jhazmat.2026.141819).

Data availability

Data will be made available on request.

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